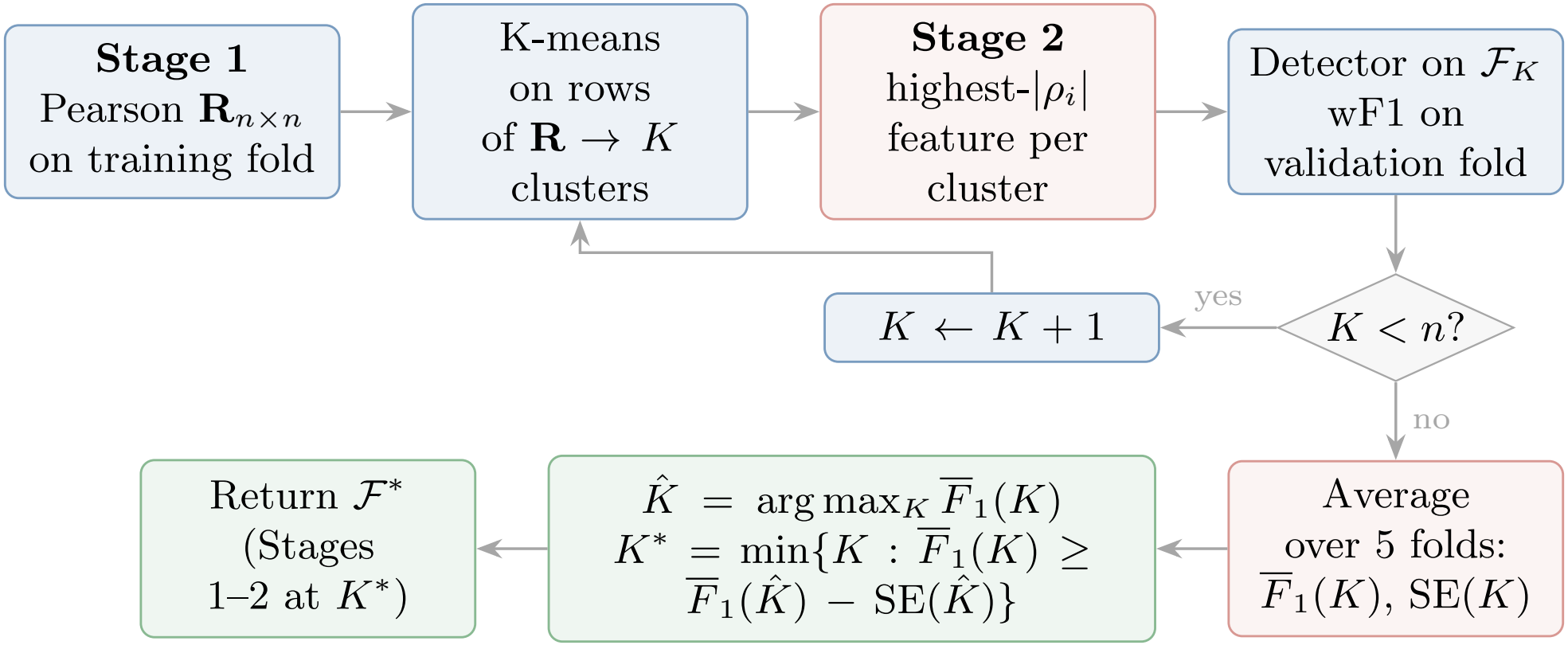

Stage 1
Pearson $\mathbf{R}_{n\times n}$
on training fold
K-means
on rows
of $\mathbf{R} \to K$
clusters
Stage 2
highest-$|\rho_i|$
feature per
cluster
Detector on $\mathcal{F}_K$
wF1 on
validation fold
$K < n$?
yes
$K \leftarrow K + 1$
no
Average
over 5 folds:
$\overline{F}_1(K)$, SE(K)
$\hat{K} = \arg\max_K \overline{F}_1(K)$
$K^* = \min\{K : \overline{F}_1(K) \geq \overline{F}_1(\hat{K}) - \mathrm{SE}(\hat{K})\}$
Return $\mathcal{F}^*$
(Stages
1–2 at K^*)